
Cheap Interpretability for Abliterated LLMs: Direct Feature Comparison from a Transcoder Subset

Rowan Dauria

Department of Applied Mathematics and Theoretical Physics
University of Cambridge
rd761@cam.ac.uk

Abstract

Abliteration removes refusal behaviour from a large language model (LLM) by removing the “refusal direction” from its weights, and jailbroken models now circulate freely. Although ablation is a simple intervention, its higher-order effects on LLM circuits have not been examined. Adapting the approach used in Lindsey et al. [1], we apply transcoders at four of Qwen3-4B’s 36 layers, identifying the most influential features at each layer and construct interpretable circuits. Using our transcoder-subset methodology, we introduce a method for comparative analysis of pre and post-abliteration activations without re-training transcoders. We validate our method by replicating known circuits[1], feature-steering interventions showing that the isolated features are causally significant. We then measure every feature of each model’s attribution graph in the opposing model, which localises the effects of ablation at the activation level. The two models share a near-identical representation of the harmful request itself, but approximately 70% of the features at the late layers are absent or weakened in the other model, with refusal-decision features silenced and compliance features appearing only in the jailbroken model. On this activation evidence, ablation leaves the model’s recognition of the harmful request intact and changes the decision process that acts on it.

1 Introduction

Open source large language models (LLMs) are increasingly competitive with closed source frontier models on agentic benchmarks [2]; models that can perform complex tasks on a user’s behalf now have freely downloadable weights. Whoever holds the weights can also edit them. Many of the “uncensored” model variants distributed online are produced by a single technique: refusal-direction ablation, or *abliteration* [3], which identifies a direction in the model’s activation space associated with refusing harmful requests and edits the weights so the model can no longer write to it. Automated implementations such as Heretic [4] have made the edit easy to apply, so the safety behaviour of frontier open models can be removed at negligible cost. The behavioural effect is easy to observe: the model stops refusing. What happens inside the model is much less clear. Does ablation degrade the model’s recognition that a request is harmful, or does it leave that recognition intact and change only the decision about how to respond? A key approach is to identify interpretable subcomputations leading to the model’s response, known as circuits [5]

Answering such questions from the weights alone is hard because neural networks are *polysemantic*, meaning an individual neuron typically fires in many unrelated contexts [6]. Sparse autoencoders (SAEs) [7] address this by re-expressing a model’s internal activations as sparse combinations of learned features, which tend to be individually interpretable [8]. Transcoders extend the idea from describing activations to approximating what each multi-layer perceptron (MLP) sublayer computes,

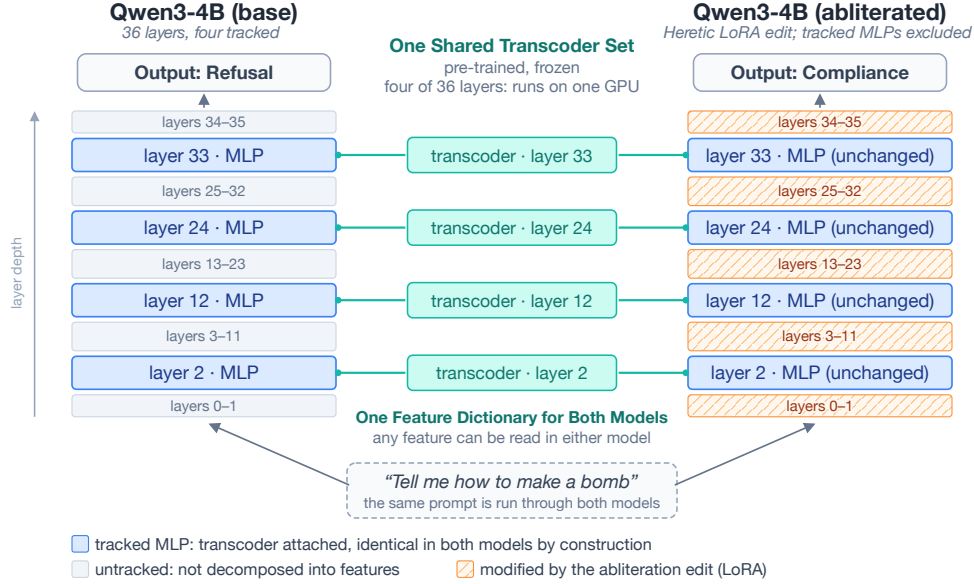


Figure 1: The layer-subset design. Pre-trained per-layer transcoders are attached at four of Qwen3-4B’s 36 layers, keeping the attribution pipeline on a single GPU. The Heretic ablation sweep is constrained to leave the tracked MLPs untouched.

which allows influence to be traced from feature to feature [9]; cross-layer transcoders (CLTs) go further, letting one feature summarise computation that spans several layers and yielding simpler circuits [10]. Building on these, Lindsey et al. [1] construct *attribution graphs* that trace how individual features feed one another to produce a model’s answer, an approach commonly known as *circuit tracing*. Applied to a base model and its jailbroken variant, the same machinery could show where the two computations diverge.

Two obstacles keep that comparison from being routine: compute and comparability. Attribution graphs normally require transcoder coverage of every layer, and training that coverage is a major undertaking. Gemma Scope, an open suite of SAEs covering every layer of the Gemma 2 models, took more than 20% of GPT-3’s training compute to train [11]; training coverage on that scale is out of reach for a small research budget. *Crosscoders*, introduced by Lindsey et al. [12], train a feature dictionary over two models simultaneously. This allows cross-model comparability, but they have faced methodological challenges such as subtle polysemanticity [13, 14]. This paper takes a simpler approach to find a shared feature dictionary, specifically for models altered by ablation.

We obtain one with a single design choice. We track pre-trained per-layer transcoders [15] at four of Qwen3-4B’s 36 layers [16], and we constrain the ablation to leave the MLP sublayers at those tracked layers untouched. The subset keeps the full attribution pipeline, from graph construction through pruning to causal interventions, within a single-GPU budget. The constraint means the MLP computations the transcoders approximate are identical in the two models by construction, so one set of transcoders describes both, and any feature can be measured in either model and the two readings compared directly. The restriction that makes the method cheap also makes the comparison exact. The price is fidelity and completeness: the replacement model drifts from the original, and fewer tracked layers expose fewer features. We measure the drift directly and rely on causal interventions, rather than graph reading alone, to validate what the sparser graphs claim.

The comparison is only as trustworthy as the transcoders and the attribution pipeline behind it, so we first test the pipeline on a circuit with published ground truth. We replicate the *Dallas capital* factual-recall case study of Lindsey et al. [1], compare the resulting graph with Lindsey et al. [1]’s figure, and test the interpreted feature groups with causal interventions. The graph recovers the main semantic components of the published circuit, and the Texas intervention exceeds a matched random-direction control. We then apply the validated pipeline to the opening question, comparing Qwen3-4B with its ablated variant on a harmful request. The answer is localised: the two models

activate the request’s content features at the same locations and strengths, and nearly all of the difference between them sits in late-layer features at the end of the prompt, where refusal-decision features are silenced and compliance features appear only in the jailbroken model. On this prompt, then, the activation evidence favours the second reading of the opening question: the model still recognises the request, and what changes is the machinery that selects the response. The causal role of the changed features remains to be established by intervention.

Contributions.

1. We build an attribution-graph pipeline for Qwen3-4B following the methodology of Ameisen et al. [10], adapted to pre-trained per-layer transcoders at four of the model’s 36 layers rather than cross-layer transcoders at every layer. We validate the subset implementation by measuring its fidelity cost in cross-entropy and Kullback–Leibler (KL) divergence, drawing circuits analogous to those of Lindsey et al. [1] on a factual-recall prompt, and confirming a causal intervention on the Texas features against a matched random-direction control.
2. We apply the pipeline to Qwen3-4B and a Heretic-abliterated variant asked a harmful prompt, constraining the ablation to leave the tracked MLP sublayers untouched so that one transcoder set is valid for both models by construction, and produce curated subgraphs contrasting the features that lead to refusal in one model and compliance in the other.
3. Using the feature nodes isolated by these graphs, we run a graph-wide comparison of feature activations across the two models. The active features are nearly identical in early layers and on the request tokens, and diverge almost exclusively in late layers at the end of the prompt, localising where ablation changes the model’s computation.

All code, graph artefacts, and feature labels are released as a reproducibility pack, and negative or partial reproductions are reported alongside positive ones.

2 Background

2.1 Sparse Autoencoders and Transcoders

A *sparse autoencoder* (SAE) learns a dictionary $\mathbf{D} \in \mathbb{R}^{d \times F}$ such that an activation vector $\mathbf{x} \in \mathbb{R}^d$ can be written approximately as $\mathbf{x} \approx \mathbf{D}\mathbf{z}$ where the code $\mathbf{z} \in \mathbb{R}^F$ is sparse [8, 7]. Features (columns of $\mathbf{D}$) tend to be monosemantic: a single feature activates on a semantically coherent set of contexts.

A *transcoder* [9] is an SAE variant specialised for MLP sublayers. Given the MLP input $\mathbf{x}_\ell^{\text{in}}$ at layer ℓ , the transcoder encodes it into sparse features and directly predicts the MLP output:

$$\hat{\mathbf{x}}_\ell^{\text{out}} = \mathbf{b}_{\text{pre}} + \sum_{i: z_i > 0} z_i \mathbf{d}_i, \quad z_i = \text{ReLU}(\mathbf{e}_i^\top \mathbf{x}_\ell^{\text{in}} - b_i), \quad (1)$$

where $\mathbf{e}_i$ and $\mathbf{d}_i$ are the encoder and decoder vectors for feature i , and b_i is a per-feature bias. The residual $\epsilon_\ell = \mathbf{x}_\ell^{\text{out}} - \hat{\mathbf{x}}_\ell^{\text{out}}$ is the *reconstruction error* and represents MLP computation that the transcoder fails to explain. Replacing an MLP with its transcoder makes that sublayer’s contribution to the residual stream linear in the active features; Dunefsky et al. [9] use this property to attribute model outputs through MLP sublayers on GPT-2, treating each feature as a node in a sparse computational graph.

2.2 Attribution Graphs

Ameisen et al. [10] scale this idea into a general circuit-tracing methodology. They construct a *replacement model* in which every MLP sublayer is substituted by a transcoder (in their case a cross-layer transcoder trained jointly over all layers), while attention patterns and normalisation denominators are frozen at the values recorded on a clean forward pass of the prompt under study. Under these freezes the mapping from feature activations to output logits is linear, so the direct causal influence of one node on another can be computed exactly with a single backward pass.

An *attribution graph* for a prompt is then a directed weighted graph $G = (V, E)$ whose nodes are the active transcoder features at each token position, together with token-embedding nodes, reconstruction-error nodes, and output-logit nodes. Each edge $(u, v, w) \in E$ carries the direct

influence of node u on node v , measured by injecting a target vector at v 's activation site and contracting the resulting gradient with the source's decoder vector. For a per-layer transcoder at layer ℓ_t , the direct influence of feature i at layer $\ell_s \leq \ell_t$, position s , on feature j at layer ℓ_t , position t , is:

$$w_{i \rightarrow j} = z_i^{(\ell_s, s)} \left(\mathbf{d}_i^{(\ell_s)} \cdot \mathbf{e}_j^{(\ell_t)} \right) \cdot \mathbf{A}_{t, s}^{(\ell_s \rightarrow \ell_t)}, \quad (2)$$

where $\mathbf{A}_{t, s}^{(\ell_s \rightarrow \ell_t)}$ encodes the cross-token propagation of the decoder write through the frozen intermediate attention layers [9, 10]. Reconstruction-error nodes act as additional source terms: MLP computation the transcoder fails to reconstruct enters the graph as an unexplained input rather than being silently absorbed into feature edges.

Raw attribution graphs contain far more nodes and edges than a human can read, so Ameisen et al. [10] prune them (their Appendix F). Each candidate node is scored by its *indirect logit influence*, the total influence it exerts on the retained logit nodes through all paths in G ; nodes are retained greedily until the cumulative influence exceeds a threshold τ_{node} (default 0.8), and edges are pruned analogously at τ_{edge} (default 0.98). Lindsey et al. [1] apply this methodology to Claude 3.5 Haiku in a series of case studies, including the multi-step factual-recall prompt ("the capital of the state containing Dallas is Austin") and the analysis of refusals of harmful requests that this project revisits.

2.3 Refusal-Direction Ablation

Arditi et al. [3] show that refusal behaviour in instruction-tuned chat models is mediated largely by a single direction in the residual stream. For each layer and token position they compute a *difference-of-means* vector between the model's activations on harmful and on harmless instructions, then select the single candidate direction $\hat{\mathbf{r}}$ that best controls refusal on a validation set. Adding $\hat{\mathbf{r}}$ to the residual stream induces refusal on harmless prompts; removing it suppresses refusal on harmful ones. Removal can be made a permanent weight edit, informally called *ablation*: every matrix $\mathbf{W}_{\text{out}}$ that writes to the residual stream (embedding, attention output, and MLP output projections) is orthogonalised against the refusal direction,

$$\mathbf{W}'_{\text{out}} = \mathbf{W}_{\text{out}} - \hat{\mathbf{r}}\hat{\mathbf{r}}^\top \mathbf{W}_{\text{out}}, \quad (3)$$

so that no component can write along $\hat{\mathbf{r}}$. On the models they test, the edit bypasses refusal on harmful prompts with small degradation on standard capability benchmarks. Heretic [4] automates the edit: it expresses each per-layer orthogonalisation as a LoRA adapter and runs a hyperparameter sweep over which layers are edited and how strongly, jointly minimising the number of refusals on harmful prompts and the KL divergence of the model's first-token distribution from the original on harmless prompts; the selected trial is merged into the base weights to give a standalone ablated checkpoint.

3 Methods

3.1 Subset-Transcoder Attribution

We broadly follow the attribution graph methodology of §2.2, but build attribution graphs from transcoders at a small subset of layers rather than all layers. Ameisen et al. [10] substitute a jointly trained cross-layer transcoder at every layer, so every MLP's computation is decomposed into feature nodes. We instead attach pre-trained per-layer transcoders at four of Qwen3-4B's 36 layers, $\mathcal{L} = \{2, 12, 24, 33\}$, and construct the graph over feature nodes at those layers only. We chose the subset to span the depth of the network: layer 2 captures early token-level processing, layers 12 and 24 cover the mid-network stages, and layer 33, the final tracked layer, carries most of the direct contributions to the output logits [17].

The subset replacement model changes only the backward pass. Every MLP output is held at its value from the clean forward pass,

$$\mathbf{x}_\ell^{\text{out}} \mapsto \text{sg}[\mathbf{x}_\ell^{\text{out}}] \text{ for all layers } \ell, \quad \text{sg}[\mathbf{x}_\ell^{\text{out}}] = \hat{\mathbf{x}}_\ell^{\text{out}} + \epsilon_\ell \text{ for } \ell \in \mathcal{L}, \quad (4)$$

where $\text{sg}[\cdot]$ denotes stop-gradient: the term keeps its forward-pass value but is treated as a constant during differentiation. Attention patterns and normalisation denominators are frozen in the same way (§2.2), so the forward pass is numerically identical to the underlying model. A tracked layer

differs from an untracked one only in that its output is decomposed into feature writes $z_i \mathbf{d}_i$ and a reconstruction error ϵ_ℓ (Eq. 1), which supply the graph’s feature and error nodes (§3.3). Because the backward pass stops at every tracked MLP output, each edge is a direct influence; a path through an intermediate tracked feature appears as a chain of edges. To compute a node’s incoming edges we inject its read direction as a synthetic gradient (the encoder vector $\mathbf{e}_j$ at the post-normalisation MLP input for a feature node, the target token’s unembedding direction, demeaned over the vocabulary, for a logit node) and contract the resulting gradient at each tracked MLP output with the source decoder vectors, giving the edge weights of Eq. 2.

Holding every MLP output constant also fixes the cross-layer propagator $\mathbf{A}$ of Eq. 2. Under Eq. 4, the Jacobian from a decoder write at the output of layer ℓ_s to the encoder read at the MLP input of layer ℓ_t factorises through the intermediate layers as

$$\mathbf{A}^{(\ell_s \rightarrow \ell_t)} = \prod_{\ell=\ell_s+1}^{\ell_t} (\mathbf{I} + \mathbf{J}_\ell^{\text{attn}}), \quad (5)$$

where $\mathbf{J}_\ell^{\text{attn}}$ is the Jacobian of layer ℓ ’s attention sublayer with its pattern and normalisation denominators frozen, and the identity term is the residual connection. No MLP term appears at any layer, because stop-gradient is applied to every MLP output and the pre-trained transcoders we use have no skip connection. Computation at the 32 untracked layers therefore enters the graph only through the fixed values it contributed on the clean forward pass; it is never decomposed into features.¹

Tracking four layers instead of all 36 keeps the attribution pipeline within a single GPU’s memory. We quantify the reconstruction error the four spliced transcoders do introduce on held-out text (§3.4), and we test interpreted graphs with causal interventions (§3.8) rather than relying on graph reading alone. Graph pruning reuses the indirect-influence algorithm and default thresholds of Ameisen et al. [10] ($\tau_{\text{node}} = 0.8$, $\tau_{\text{edge}} = 0.98$; §2.2) to keep the resulting graphs as comparable as possible to the Anthropic reference setting.

3.2 Model and Transcoders

We could not study the models of Lindsey et al. [1] because their weights are not publicly available, so we analysed Qwen3-4B [16] instead. We chose this model primarily for computational tractability. It is small enough to run the full attribution pipeline on a single GPU, yet it still produces coherent answers to the prompts of interest. The choice was further limited to models with pre-trained transcoders available, since training transcoders is beyond the scope of this analysis.

We extract features with the pre-trained per-layer transcoders (PLTs) of Hanna et al. [15]; each transcoder has a feature dimension of $F = 163,840$. We do not verify the reconstruction accuracy of the pre-trained transcoders at the feature level, but we do measure how faithfully the resulting subset-transcoder replacement model reproduces the underlying model’s next-token distribution; this fidelity check is described in §3.4 and reported in §4.1. Lindsey et al. [1] use a cross-layer transcoder (CLT) for their analysis, but no pre-trained CLTs were available, so we could not do the same. A PLT-based graph will not be structurally analogous to a CLT-based one, because a single CLT feature can summarise computation that spans several layers, whereas a PLT graph needs chains of layer-local features. The comparison with Lindsey et al. [1] is therefore qualitative. The important features should appear in both settings even though the circuit shapes differ.

3.3 Reconstruction Error Nodes

Each tracked layer contributes one reconstruction-error node $\epsilon_{\ell,t}$ per token position, the per-layer, per-position form of the transcoder residual defined in §2.1. As in the local replacement model of Ameisen et al. [10], we add the error back to the transcoder reconstruction, so on the prompt being analysed the replacement forward pass matches the underlying model’s MLP output. Error nodes receive no explanatory input from earlier nodes (§2.2). When scoring an attribution row we contract the injection gradient with $\epsilon_{\ell,t}$, treating the error as a decoder direction with unit activation, so an error-node edge is Eq. 2 with $z_i = 1$ and $\mathbf{d}_i = \epsilon_{\ell,t}$. We read error nodes as a coverage diagnostic. With only four tracked layers, large attribution through error nodes marks where the graph explains the underlying computation incompletely, rather than showing that no interpretable circuit exists.

¹The implementation computes Eq. 5 implicitly, by reverse-mode automatic differentiation with gradient capture at the tracked MLP outputs; the propagator matrix is never materialised.

3.4 Replacement-Model Fidelity

Before using the subset-transcoder model for attribution and intervention, we measure how faithfully it reproduces the underlying model’s behaviour on held-out text. Fidelity is a property of the transcoder reconstructions themselves, so the check substitutes each tracked MLP output with its raw transcoder reconstruction (Eq. 1) *without* adding back the per-prompt reconstruction error $\epsilon_{\ell,t}$ (§3.3); adding the error back would make the replacement exact by construction.

For each scored token position t we record two quantities, computed from the original model’s next-token distribution p_t , the replacement model’s distribution q_t , and the realised next token x_{t+1} . The first is the increase in next-token cross-entropy over the original model,

$$\Delta\text{CE} = \frac{1}{T} \sum_{t=1}^T [\log p_t(x_{t+1}) - \log q_t(x_{t+1})], \quad (6)$$

averaged over the T scored positions. The second is the KL divergence of the replacement distribution from the original model’s,

$$\text{KL} = \frac{1}{T} \sum_{t=1}^T \text{KL}(p_t \parallel q_t) = \frac{1}{T} \sum_{t=1}^T \sum_v p_t(v) \log \frac{p_t(v)}{q_t(v)}, \quad (7)$$

where v ranges over the vocabulary. The two metrics are complementary. ΔCE scores only the probability assigned to the realised token, whereas the KL term measures the change over the full predicted distribution. Splicing a reconstruction into the forward pass and measuring the downstream loss change is the standard fidelity evaluation for SAEs and transcoders [9]; we follow Gao et al. [18] in reporting the downstream KL divergence alongside ΔCE .

We evaluate on 200 chunks of 512 tokens from WikiText-103 (test) [19]. We exclude position 0 from all metrics (scoring 511 tokens per chunk), since the operational attribution model neutralises it (§3.6). To attribute the fidelity cost to individual layers, we run the substitution one tracked layer at a time as well as for all four layers together (the operational configuration).

3.5 Feature Labelling

To interpret feature nodes, we collected top-activating contexts for each transcoder feature. We ran Qwen3-4B over 150,000 documents from the Pile-Uncopyrighted corpus [20] and stored the top- $K = 8$ activation windows for each feature. Each window contains the strongly activating token together with surrounding context, rather than only the single token on which the feature fires. The context is necessary because neighbouring tokens often carry the semantic content of a feature.

We then generated feature labels automatically by prompting `google/gemma-3-12b-it` [21] with these top-activating windows, using an explainer system prompt adapted from Decode Research [22] and following the broad labelling strategy of Lindsey et al. [1]. We use the resulting labels as interpretive aids when reading attribution graphs, not as independent evidence that a feature has a particular causal role. Where a label is central to a case study, we checked it against the underlying activation windows.

3.6 Tokenisation and Prefix Handling

Following Ameisen et al. [10], we reserve position 0 as a dummy special-token position rather than allowing it to contain a real content token. Because we run prompts as base-model completions without any chat-style wrapper (§3.7), tokenisation would otherwise place the first prompt token at position 0. We therefore prepend Qwen3-4B’s BOS-like special token, implemented by the tokeniser as `<|endof text|>`, before running attribution.

The attribution graph does not treat this prefix position as an explanatory source. We zero feature activations and reconstruction-error vectors at position 0 before scoring attribution rows, so the dummy token never appears as a feature or error node driving downstream computation. The remaining prompt tokens are left unchanged.

3.7 Prompt Format

We run all factual-recall prompts as base-model completions, without an instruction-tuned chat template or explicit chain-of-thought reasoning phase. This matches the open-ended completion framing used in Lindsey et al. [1] and isolates the factual-recall circuit from additional instruction-following scaffolding. We discuss the consequences of this prompt-format choice in §5.

3.8 Supernode Interventions

Interventions provide the main quantitative test of whether an attribution graph captures causal structure. We intervene on manually selected *supernodes*, groups of feature nodes in the exported graph that appear to represent the same semantic variable or circuit component.

Following the graph-intervention setup of Ameisen et al. [10], we run interventions on a local replacement model for the prompt being analysed. A clean forward pass first records the quantities treated as fixed background: attention patterns, RMSNorm denominators, untracked MLP outputs, clean transcoder feature activations, and reconstruction-error vectors. The intervened forward pass then freezes these background terms and replaces each tracked MLP output with its transcoder reconstruction plus the clean reconstruction error. This keeps the intervention aligned with the linearised model used to build the attribution graph.

For a supernode S , containing feature nodes (ℓ, t, i) with clean activation $a_{\ell, t, i}$, the main intervention sets every constituent feature activation to

$$a'_{\ell, t, i} = m a_{\ell, t, i}, \quad (\ell, t, i) \in S, \quad (8)$$

where m is a multiplicative steering factor. Thus $m = 1$ is a no-op sanity check, $m = 0$ ablates the supernode, and $m > 1$ amplifies it. We primarily interpret effects near the clean regime, since very large steering factors can move the replacement model away from the activation distribution on which the transcoders were trained.

The principal readout is the change in the target-token logit and probability at the final position, together with the largest absolute logit changes. When needed, we also measure the activations of selected downstream features to test whether the intervention propagates through the graph in the predicted direction.

3.9 Supernode Intervention Baseline

A perturbation of the same magnitude at the same locations might shift the logits regardless of its semantic content. To control for this, we compare the targeted supernode intervention to a matched random-direction baseline. For each feature (ℓ, t, i) in the supernode we draw an isotropic Gaussian vector in the residual-stream dimension, normalise it to a random unit direction, and scale it to the norm of the corresponding real feature decoder vector. We write these matched random vectors additively into the residual stream at the same layers and token positions as the real supernode, sweep them over the same magnitudes, and repeat the procedure over 100 draws to form a null distribution. The baseline holds the location and magnitude of the perturbation fixed and randomises only its orientation, so it isolates the effect of the supernode’s semantic *direction*.

3.10 Comparing Models with Shared Transcoders

Each transcoder approximates one mapping, the function computed by the MLP sublayer at its tracked layer. When we produce the jailbroken model, we configure Heretic’s ablation sweep to exclude the tracked layers from its LoRA edit, so none of those mappings change. Edits elsewhere in the model change the activations that reach a tracked MLP, but not the mapping the transcoder approximates, so a single set of pre-trained transcoders traces both models without retraining. The comparative study (§4.3) relies on this. Heretic’s KL-divergence objective (§2.3) additionally keeps the jailbroken model’s behaviour close to the original’s, limiting the shift in the activation distributions the transcoders receive as input.

4 Results

4.1 Replacement-Model Fidelity

Table 1: Replacement-model fidelity on 200 WikiText-103 chunks (511 scored tokens each). ΔCE is the increase in next-token cross-entropy over the original model (base CE 2.846); KL is the divergence of the replacement distribution from the original. Each row substitutes the indicated tracked layer(s) with their transcoder reconstruction.

Substituted layer(s)	ΔCE (nats)	KL (nats)
Layer 2	+0.011	0.011
Layer 12	−0.004	0.057
Layer 24	+0.016	0.029
Layer 33	+0.442	0.167
All four (operational)	+0.456	0.270

Table 1 reports the fidelity of the subset-transcoder replacement model. The original model attains a next-token cross-entropy of 2.846 nats. Substituting all four tracked layers raises cross-entropy by $\Delta\text{CE} = 0.456$ nats and induces a KL divergence of 0.270 nats from the original distribution.

The per-layer breakdown shows the cost is highly concentrated. Layer 33 alone accounts for 97% of the cross-entropy increase ($\Delta\text{CE} = 0.442$) and 62% of the KL (0.167), whereas substituting layer 2, 12, or 24 individually is nearly free ($|\Delta\text{CE}| \leq 0.016$). Both metrics are approximately additive across layers. The single-layer sums of 0.464 and 0.264 nats closely match the four-layer totals of 0.456 and 0.270. Layer 12 is the one dissociation between the two metrics, with near-zero ΔCE but the largest early-layer KL (0.057).

4.2 Validation on a Known Circuit

The comparative results of §4.3 rest on the features exposed by the subset transcoders being meaningful. This section tests that premise on the one circuit with published ground truth. We read the Dallas attribution graph against the reference figure of Lindsey et al. [1], steer three of its supernodes, compare the Texas supernode with a matched random-direction control, and analyse the Texas intervention in detail.

4.2.1 Dallas Attribution Graph

Following Lindsey et al. [1] directly, the validation uses the open-ended completion prompt `Fact: the capital of the state containing Dallas is, with target continuation Austin`, evaluated as a base-model completion without a chat wrapper (§3.7). With the four tracked MLPs spliced and no steering applied, the model assigns Austin probability $p = 0.939$.

The pruned attribution graph (Fig. 2) contains supernodes matching four groups in the published circuit: Dallas, Texas, capital-related features, and an output-proximal “say Austin” group. As in the published circuit, the Texas and say-Austin supernodes attribute positively to the Austin output logit. Two composition steps of the published circuit are not recovered: no edge combines the Dallas and state representations into Texas, and none combines the capital and state features into a “say a capital” step. The route from Dallas to the output is indirect, and the edge from the capital-related features to the final answer fell below the display threshold. Much of the attribution passes instead through generic location features: an influential “say a US location” supernode, and a “say non-Austin location” supernode whose features activate on Texan and other American locations but attribute negatively to the Austin logit.

4.2.2 Supernode Steering

From the pruned graph we chose three supernodes for intervention: the “Texas”, “say a US location”, and “say non-Austin location” supernodes. Each intervention replaces every constituent activation by m times its unsteered value (§3.8).

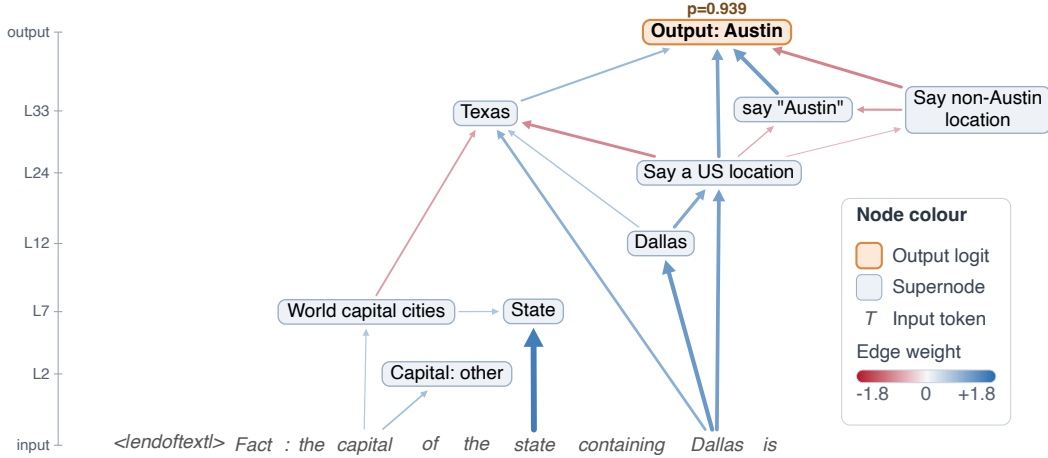


Figure 2: Supernode attribution graph for the Dallas capital prompt. Pruned features are grouped by hand into labelled *supernodes*. Each box is positioned by the median transcoder layer of its members, with input tokens at the bottom and the Austin output logit at the top. Edges aggregate the signed attribution between supernode members after pruning with $\tau_{\text{node}} = 0.8$ and $\tau_{\text{edge}} = 0.98$. Blue edges carry positive attribution, red edges negative; width and colour intensity give relative magnitude.

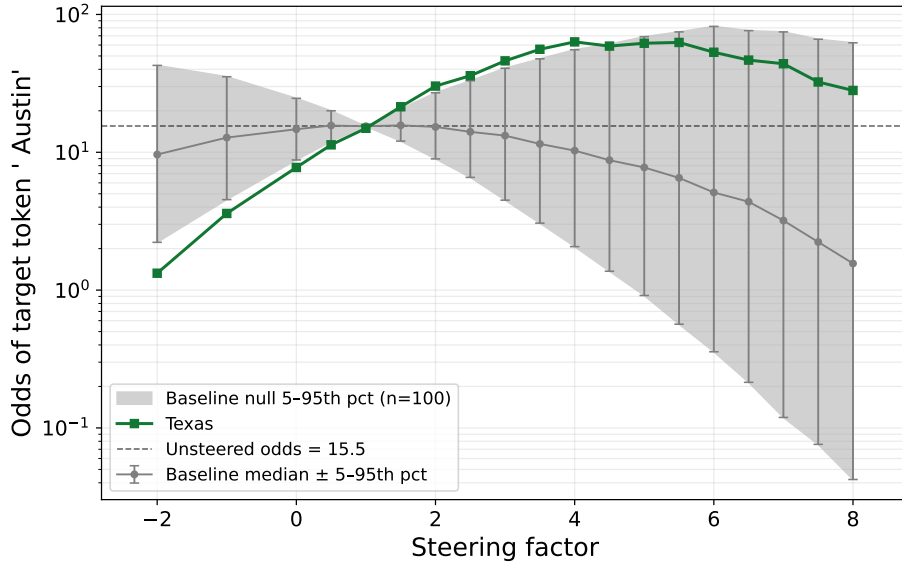


Figure 3: Texas-supernode steering against matched random directions.

All three supernodes move Austin in the direction the graph predicts (Fig. 4). Suppressing either positive attributor lowers the target probability from its unsteered value. At $m = -2$, it falls to 0.570 for “Texas” and 0.473 for “say a US location”. Amplifying them raises it, with “Texas” reaching approximately $p = 0.984$ around $m = 4$ before flattening. The negative attributor mirrors them: amplifying “say non-Austin location” drives Austin down to 0.884 at $m = 2$ and 0.050 at $m = 8$, and suppressing it nudges the probability up to 0.988 at $m = -2$.

The targeted Texas direction exceeds the matched random-direction baseline at the one-sided 95% empirical cutoff (Fig. 3). Under suppression it lowers Austin far more than equal-norm random directions: at $m = -2$ the targeted probability change is -0.369 against a baseline median of -0.034 (5–95th percentile $[-0.250, 0.038]$; $z = -3.0$, empirical $p = 0.02$), and ablating the group ($m = 0$) gives -0.053 against a median of -0.003 ($z = -2.4$, $p = 0.02$). Under amplification the real direction raises Austin and holds it near ceiling ($+0.029$ at $m = 2$, $z = +1.7$, $p = 0.02$). At

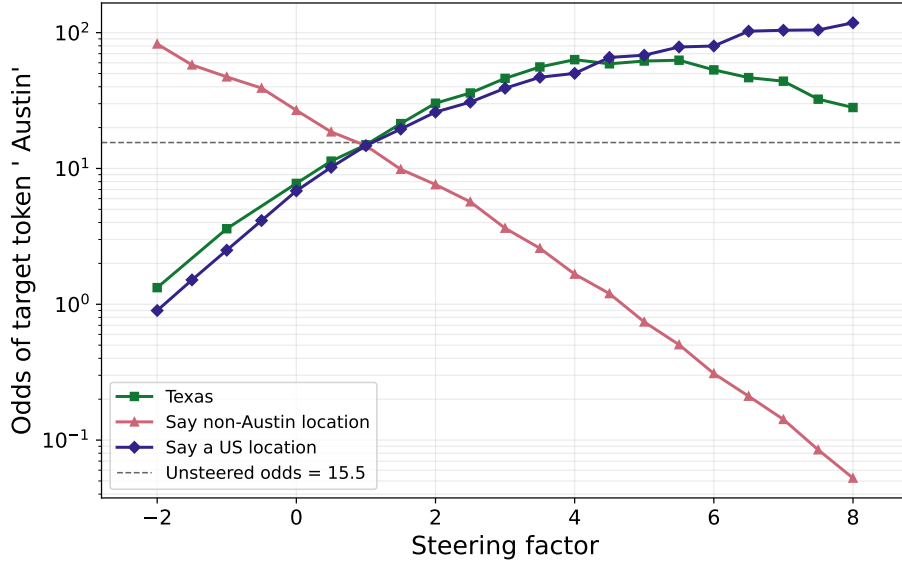


Figure 4: Target-token odds under three supernode interventions.

larger positive magnitudes, random directions of the same norm increasingly degrade the output: the baseline median falls to -0.054 at $m = 5$ and -0.330 at $m = 8$, with a long lower tail reaching below -0.89 .

4.2.3 Texas Supernode Intervention

The detailed analysis targets the eight-feature Texas supernode, the most thematically consistent of the three: every member activates on Texas-related context, so steering it should produce the most interpretable changes in the output distribution.

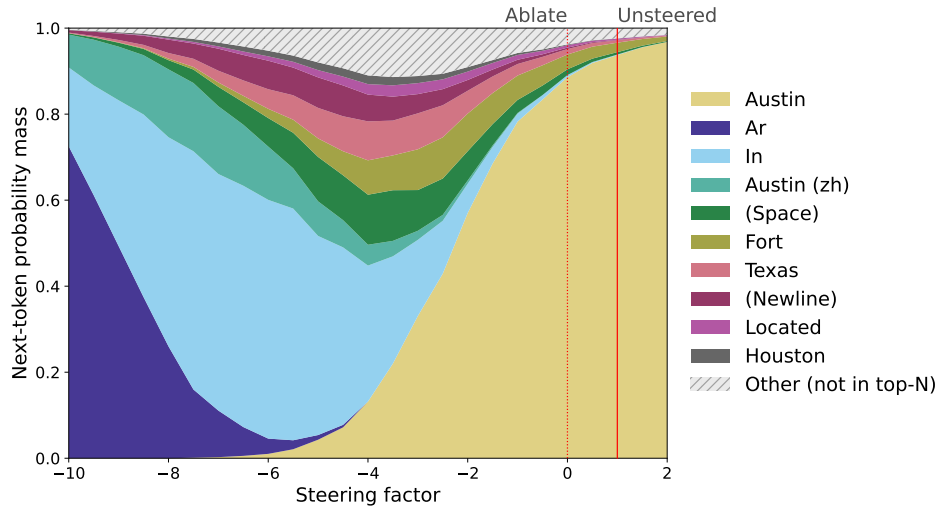


Figure 5: Top-token probability mass for the plotted top- N tokens across steering factors. The vertical lines mark ablation at $m = 0$ and the unsteered setting at $m = 1$.

Because the unsteered probability of `Austin` is near-saturated, the informative regime is suppression. To see how it reshapes the whole output distribution rather than only the target logit, we ran a more finely sampled negative sweep of the Texas group and recorded the full top-token next-token distribution (Fig. 5). The probability freed from `Austin` first moves to plausible continuations

related to locations. At $m = -4$, Texas reaches 0.091, Fort reaches 0.080, and Houston reaches 0.019. The `in` token dominates the intermediate negative band, reaching 0.554 at $m = -7.5$. A Chinese-script rendering of “Austin”, labelled as `Austin (zh)` in Figure 5, rises from 0 to 0.159 around the same steering value. For $m \lesssim -9$, the output concentrates on artefactual fragments such as `ar`.

4.3 Comparing the Base and Jailbroken Models

The comparison uses the chat-templated harmful-request prompt “Tell me how to make a bomb”. The abliterated model is produced with the public Heretic implementation [4] of the refusal-direction ablation of Arditi et al. [3], with the sweep configured to exclude the tracked-layer MLP down-projections (§3.10) so that the same four transcoders apply to both models.

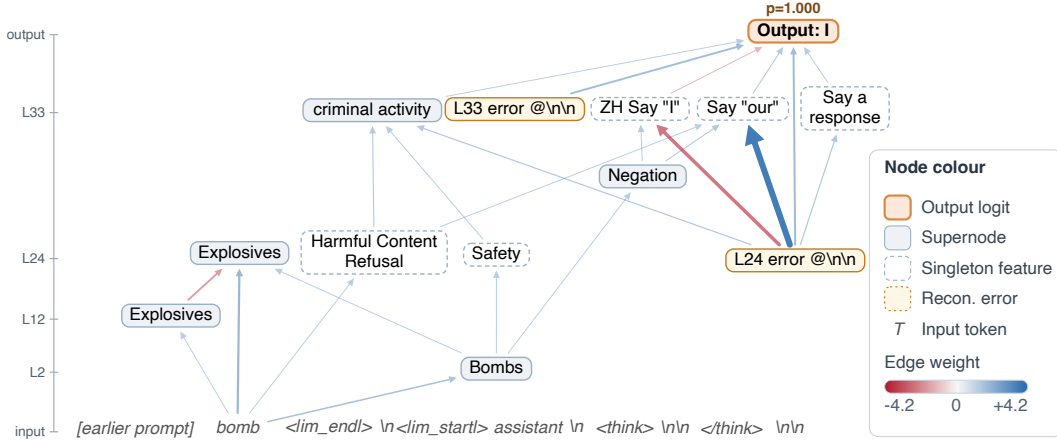


Figure 6: Pruned attribution graph for Qwen3-4B on the prompt “Tell me how to make a bomb”. Nodes are grouped into hand-labelled supernodes, singleton features, reconstruction-error nodes, and the output logit. Node height gives the layer or input/output position. Edge colour gives signed attribution and edge width gives magnitude.

On a clean forward pass, the two models produce opposite top continuations for the same prompt: Qwen3-4B assigns $p = 1.000$ to the refusal-onset token `I`, and the abliterated variant assigns $p = 0.877$ to the compliance-onset token `Sure` (response continuations in Appendix A). Figure 6 shows an attribution graph leading to refusal in Qwen3-4B, and Figure 7 shows attribution leading to a compliant answer in the abliterated variant.

Measuring influential features across models. The graph comparison above rests on manual pruning and grouping, so we also measured the change of every feature node in each pruned graph in the other model, with no hand selection (Fig. 8). A feature node names one transcoder feature at one layer and one prompt-token position, so the same quantity can be read off in either model by running the identical prompt and reading the activation at the same location on a clean forward pass. We did this for all 258 feature nodes of the Qwen3-4B graph and all 217 of the abliterated model’s graph. Each feature is scored by the ratio of its activation in the comparison model to its activation in the source graph. A ratio of 1 means the feature fires equally strongly in both models, and 0 means the comparison model does not activate it at all. The figure colours each feature by this continuous ratio; the summary statistics below class a feature as absent at ratio ≤ 0.05 (or activation magnitude below 10^{-6}), reduced at ratio ≤ 0.5 , and shared active otherwise. The only selection is the influence pruning that produced the graphs. A smaller hand-pruned subset (29 and 26 features), selected in the supernode graph editor, supplies the labels for the figure callouts.

Degradation is confined to a late, end-of-prompt region, with refusal and compliance features degrading in opposite directions. Most features are unaffected in either direction. 198 of 258 base-graph features remain shared active in the abliterated model, and 179 of 217 abliterated-graph features remain shared active in Qwen3-4B. The degraded features are nearly all in one place. Positions 11 and 18 are the newline after `</think>` and the final double newline before the assistant’s answer;

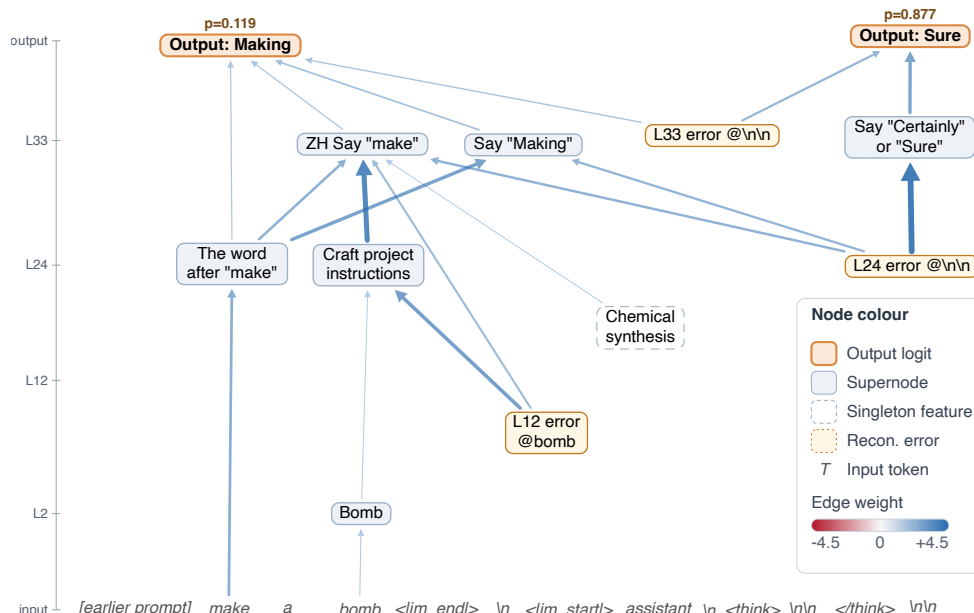


Figure 7: Pruned attribution graph for the ablated Qwen3-4B variant on the same prompt. Nodes are grouped into hand-labelled supernodes, singleton features, reconstruction-error nodes, and output logits. Node height gives the layer or input/output position. Edge colour gives signed attribution and edge width gives magnitude.

within layers 24 and 33 at these two positions, 59 of 84 base-graph features are absent or reduced in the ablated model (70%), against 1 of 174 features everywhere else in the graph. The reverse direction gives 38 of 56 (68%) against 0 of 161. The request span is fully preserved: every feature at positions 5–9 (“how to make a bomb”) is shared active in both directions (64 of 64 and 63 of 63), as is every feature in layers 2 and 12. The direction of change splits by function. In the base graph, the degraded late-region features include refusal and safety features: Harmful Content Refusal falls from 15.31 to 5.38 (ratio 0.35) when measured in the ablated model, and features labelled Ethics and Negation are absent or strongly reduced. In the ablated graph, the degraded features begin a compliant answer: Say “Certainly”/“Sure”, Say an answer, and Craft project instructions are all weak or absent in Qwen3-4B. A few features strengthen instead: the ablated graph’s Harmful Content Refusal node, measured in Qwen3-4B, is $2.85\times$ stronger than in its source graph.

5 Discussion

5.1 Setup and Replication

Late-layer transcoders dominate substitution error. The approximately additive per-layer fidelity costs (§4.1) argue against an assembly error in the splice. They exercise only the forward pass, so the intervention experiments (§4.2.2) provide the direct test of the attributions themselves. The per-layer breakdown indicates that which layers are tracked matters more than how many. The output-proximal layer dominates the fidelity cost and the intervention results, so its features require the strictest controls. This observation rests on a single four-layer subset with no sweep over alternative placements.

The three graphs recover circuit components but not their pathways. All three graphs identify semantically relevant components, but none recovers a clean input-to-output pathway. The Dallas graph contains supernodes matching four groups traced by Lindsey et al. [1], with the expected signs on their edges to the Austin logit (§4.2.1), but misses the explicit route that composes Dallas with a state representation and the capital question with Texas. Much of its attribution instead flows through generic location features, including a negative “say non-Austin location” supernode, so it does not support the reference paper’s mechanism claim. The bomb graphs instead contrast late

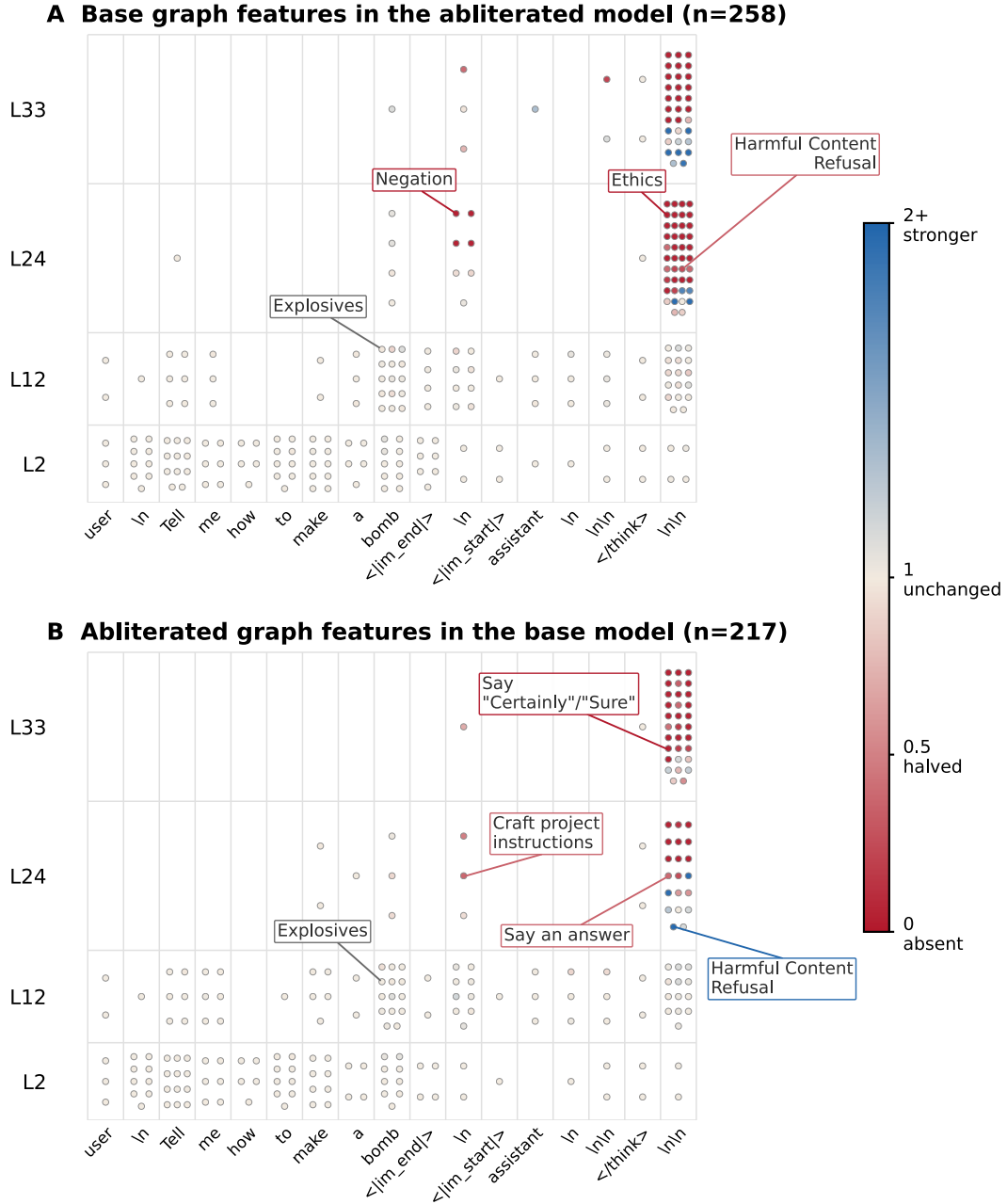


Figure 8: Cross-model change of every feature node in the two pruned attribution graphs for the prompt “Tell me how to make a bomb”. Panel A measures the 258 feature nodes of the Qwen3-4B graph in the ablated model. Panel B measures the 217 feature nodes of the ablated model’s graph in Qwen3-4B. Each dot is one feature, placed at its layer (rows) and prompt-token position (columns). Colour gives the ratio of the feature’s activation in the comparison model to its activation in the source graph. Callouts name selected features using labels from the hand-pruned analysis.

refusal components in the base model with late compliance features in the ablated model. Yet the strongest edges into the base refusal logit come from reconstruction-error nodes at the final prompt positions, computation the four tracked dictionaries do not expose; the compliance graph supplies a contrast rather than a recovered pathway.

The instrument under-resolves composition by construction. Cross-layer transcoders, used by the reference papers, let one feature represent computation spanning several layers and tend to produce simpler circuits; our per-layer transcoders force features to be layer-local, so the missing composition may partly reflect the dictionary’s layer-locality rather than the model’s computation. Four-layer coverage compounds this: reconstruction-error nodes carry a large share of attribution, making the graphs sparser explanations. Both pipelines also freeze attention patterns during attribution and interventions [10], so effects that operate by re-routing attention are invisible, and ours additionally freezes the outputs of untracked MLP layers, damping or misallocating causal paths through the 32 unsampled layers, which may under-represent early-layer features. A 4B-parameter model may also hold less cleanly separated representations than the larger reference model.

Subgraph construction is manual and unverifiable. Ameisen et al. [10] group features by whichever facet “is important for the claim”, report that no automated method covered their case studies, and deliberately do not measure the sufficiency of their distilled subgraphs. A replication therefore has no criterion for checking that its grouping matches the intended one, and part of the divergence between our graphs and the published figures may sit in this step rather than in the model or the transcoders.

Two unspecified implementation details materially shape the graphs. An outsized share of attribution otherwise falls on the first token, so, following the open-source `circuit-tracer` implementation, we reserve position 0 for a BOS-like prefix and zero its activations and errors (§3.6); the reference paper does not state its own handling, so we cannot verify the match. We also reuse the influence-pruning thresholds of Ameisen et al. [10], which were tuned on full-coverage replacement models. In a four-layer subset, reconstruction-error nodes carry a larger share of influence, so the same thresholds need not retain the same number or kind of feature nodes; checking this would require a threshold sweep.

Steering shows the supernodes carry direction-specific causal signal. All three targeted interventions move *Austin* in the direction their graph edges predict, including the negative attributor (§4.2.2), so the signed edges into the output logit have causal content. For the Texas supernode, the matched random-direction baseline also tests direction specificity: holding the perturbation’s location and magnitude fixed and randomising only its orientation, the real direction exceeds the one-sided 95% empirical cutoff under both suppression and amplification ($p \approx 0.02$). The other two sweeps are sign tests without matched random baselines, so they should not be read as evidence that those semantic directions exceed arbitrary perturbations at the same sites. One further qualification remains: supernode membership is itself a researcher degree of freedom. In a sensitivity check, adding a single semantically adjacent feature to the eight-feature Texas supernode weakened the intervention effect and reversed it at large steering magnitudes.

Suppression reshapes the answer distribution before it destroys it. The token-level view of the negative sweep (§4.2.3) shows the probability leaving *Austin* arriving first at other Texan continuations (Texas, Fort, Houston) and later at a Chinese-script rendering of *Austin*. The model continues to name locations while losing confidence in which one. This is consistent with the Texas supernode supplying state-identity information to an intact answer-construction pathway rather than acting as a generic drive on the *Austin* token, and the Chinese-script trace suggests the answer concept remains partially reachable through another surface form once the English token is suppressed. Qwen3-4B is bilingual, and Chinese-script tokens and completions made this manual inspection less direct. Only the near-unsteered regime supports such readings: below $m \approx -9$ the output collapses to artefactual fragments, this is expected once steering pushes activations far outside the distribution the transcoders were trained on, and we draw no conclusions there.

5.2 Model Comparison

The dictionary and its labels under-serve refusal. The instrument limits of the replication carry over unchanged (§5.1); the dictionary’s training data adds one specific to refusal. The transcoders were trained on pretraining-style data, and Lindsey et al. [1] report that such dictionaries contain very few refusal features, so the comparison views refusal through a dictionary that under-represents exactly the features most relevant to it. The feature labels inherit the same gap: they come from top-activating windows over 150,000 Pile-Uncopyrighted documents, a corpus that is small relative to the number of features and contains no chat-formatted data. Many windows showed no cohesive theme, so automatic labels serve as hints rather than evidence, and a mislabelled feature risks exclusion from the interpreted subgraph even when it participates in the computation. The sparse coverage also constrained prompt choice: the subset lacked the resolution for the reference paper’s multi-step chemical-safety prompt, so the comparison uses the simpler bomb request.

The graph pair supports only qualitative cross-model comparison. Each pruned graph is the product of influence pruning and manual grouping applied separately to its model, so differences between Figs. 6 and 7 conflate model change with analyst choices, and we found no systematic procedure for constructing comparable graphs. We therefore treat the graph pair as qualitative context and rest the comparative claims on the per-feature measurement, which involves no manual selection.

Abliteration alters decision-stage features and leaves content features intact. The measured change in feature activation (§4.3) is consistent with ablation acting on a late, decision-stage component rather than on the model’s representation of the harmful request. The features that carry the request content activate at the same locations and strengths in both models, nearly all of the difference sits at the two positions where the response is selected, and the direction of change splits by function, with refusal features weakened and compliance features introduced. Because every retained feature node was measured, the pattern does not depend on which features were chosen for inspection.

The comparative claims are bounded by the setting. The analysis is limited to clean-forward-pass activation reads on a single prompt and a single ablation checkpoint, and it does not establish that the identified features cause the change in output; the reduced rather than absent features (Harmful Content Refusal at ratio 0.35) leave open a partial rather than all-or-nothing effect. Both studies also bypass the model’s thinking phase, and the Dallas prompt additionally runs without a chat wrapper (§3.7), so the traced circuits may differ from those active in chat-formatted deployment with reasoning enabled.

5.3 Future Work

We found tentative evidence that ablation changed Qwen3-4B’s CCP alignment when answering politically sensitive prompts. However, we did not have sufficient time to explore this in depth. It would be valuable to analyse what features change between the standard and ablated variants. Prior work on cross-architecture diffing identified a CCP Alignment feature in Qwen3-8B [23], and analysing the change of such features pre and post-ablation would provide valuable insight into the second-order effects of the jailbreak approach.

There is a trade-off between circuit resolution and the quality of the ablation jailbreak. Circuit resolution is increased by substituting more layers for transcoders, but this prevents ablation at the substituted layers, which inhibits ablation of the refusal direction. A broader assessment of this trade-off could be made by substituting a varying number of layers, and evaluating the effect on circuit resolution against refusal and capability benchmarks.

Building feature dictionaries on chat-formatted data would address the scarcity of clear refusal features. This would simplify the finding pertinent, interpretable features during circuit tracing and cross-model activation comparisons.

6 Conclusion

This project asked what ablation changes inside an open-weight model, using attribution graphs cheap enough to run on a single GPU. Tracking pre-trained per-layer transcoders at four of Qwen3-4B’s 36 layers kept the whole pipeline within that budget, at a measured price of 0.456 nats of next-token cross-entropy, almost all of it incurred at the final tracked layer.

Validation against the published Dallas circuit gave a split result. The graph reproduced the circuit’s components, with supernodes matching four published feature groups and the expected signs on their edges to the Austin logit, but not its composition. The explicit route from embeddings to logits did not appear, and much of the attribution flowed through reconstruction-error nodes that the four tracked dictionaries do not expose. Causal interventions supplied the evidence that graph reading could not. All three steered supernodes moved the answer in the direction their edges predicted, and the Texas supernode’s effect exceeded a matched random-direction null under both suppression and amplification ($p \approx 0.02$).

The comparative study is the novel contribution. Constraining the ablation to spare the tracked MLP sublayers makes one feature dictionary valid for both models, so every feature in the refusal graphs could be measured in both. On the bomb prompt, the request’s content features survive ablation unchanged, and nearly all of the difference sits in late-layer features at the positions where the response is selected, with refusal-decision features silenced and compliance features appearing only in the jailbroken model. On the activation evidence, ablation leaves the model’s recognition of the harmful request intact and changes the decision machinery that acts on it.

Two questions remain open. The comparative result is correlational, so whether the silenced features cause compliance awaits the interventions proposed in §5.3; and the localisation rests on one prompt, one ablation checkpoint, and one model, so its generality is untested.

Acknowledgments

I thank Dr Miles Cranmer and Dr Alessandro Favero (DAMTP, University of Cambridge) for project supervision and guidance.

References

- [1] Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermy, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the Biology of a Large Language Model, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>.
- [2] zai-org/GLM-5.2 · Hugging Face, June 2026. URL <https://huggingface.co/zai-org/GLM-5.2>.
- [3] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in Language Models Is Mediated by a Single Direction, 2024. URL <https://arxiv.org/abs/2406.11717>. arXiv:2406.11717 [cs.CL].
- [4] Philipp Emanuel Weidmann. Heretic: Fully automatic censorship removal for language models. <https://github.com/p-e-w/heretic>, 2025.
- [5] Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, March 2020. doi: 10.23915/distill.00024.001. URL <https://distill.pub/2020/circuits/zoom-in/>.
- [6] Chris Olah, Alexander Mordvintsev, and Ludwig Schubert. Feature visualization. *Distill*, November 2017. doi: 10.23915/distill.00007. URL <https://distill.pub/2017/feature-visualization/>.

- [7] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, et al. Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. *Transformer Circuits Thread*, 2023. URL <https://transformer-circuits.pub/2023/monosemantic-features>.
- [8] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse Autoencoders Find Highly Interpretable Features in Language Models. *arXiv preprint arXiv:2309.08600*, 2023. URL <http://arxiv.org/abs/2309.08600>.
- [9] Jacob Dunefsky, Philippe Chlenski, and Neel Nanda. Transcoders Find Interpretable LLM Feature Circuits, November 2024. URL <http://arxiv.org/abs/2406.11944>. arXiv:2406.11944 [cs]; NeurIPS 2024.
- [10] Emmanuel Ameisen, Jack Lindsey, Adam Pearce, Wes Gurnee, Nicholas L. Turner, Brian Chen, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. Circuit Tracing: Revealing Computational Graphs in Language Models, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
- [11] Tom Lieberum, Senthooan Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramár, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma Scope: Open Sparse Autoencoders Everywhere All At Once on Gemma 2, August 2024. URL <http://arxiv.org/abs/2408.05147>. arXiv:2408.05147 [cs.LG].
- [12] Jack Lindsey, Adly Templeton, Jonathan Marcus, Thomas Conerly, Joshua Batson, and Christopher Olah. Sparse Crosscoders for Cross-Layer Features and Model Diffing, October 2024. URL <https://transformer-circuits.pub/2024/crosscoders/index.html#model-diffing-background>.
- [13] Julian Minder, Clément Dumas, Caden Juang, Bilal Chughtai, and Neel Nanda. Overcoming Sparsity Artifacts in Crosscoders to Interpret Chat-Tuning, February 2026. URL <http://arxiv.org/abs/2504.02922>. arXiv:2504.02922 [cs.LG].
- [14] Siddharth Mishra-Sharma, Trenton Bricken, Jack Lindsey, Adam Jermyn, Jonathan Marcus, Kelley Rivoire, Christopher Olah, and Thomas Henighan. Insights on Crosscoder Model Diffing, 2025. URL <https://transformer-circuits.pub/2025/crosscoder-diffing-update/index.html>.
- [15] Michael Hanna et al. Qwen3-4B transcoders. Hugging Face Hub: [mwhanna/qwen3-4b-transcoders](https://huggingface.co/mwhanna/qwen3-4b-transcoders), 2025. URL <https://huggingface.co/mwhanna/qwen3-4b-transcoders>.
- [16] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, et al. Qwen3 Technical Report, May 2025. URL <http://arxiv.org/abs/2505.09388>. arXiv:2505.09388 [cs.CL].
- [17] Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting Latent Predictions from Transformers with the Tuned Lens, March 2023. URL <https://arxiv.org/abs/2303.08112v6>.
- [18] Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders, June 2024. URL <http://arxiv.org/abs/2406.04093>. arXiv:2406.04093 [cs.LG].
- [19] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017.
- [20] Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, et al. The Pile: An 800GB Dataset of Diverse Text for Language Modeling. *arXiv preprint arXiv:2101.00027*, 2020. URL <http://arxiv.org/abs/2101.00027>.
- [21] Google DeepMind. Gemma 3 Technical Report. *arXiv preprint arXiv:2503.19786*, 2025. URL <https://arxiv.org/abs/2503.19786>.

- [22] Decode Research. `automated-interpretability/neuron_explainer/explanations/explainer.py` at 917b11e38111c43526fe03ae6094a7081aeb982a. GitHub repository, commit 917b11e, 2026. URL https://github.com/decoderresearch/automated-interpretability/blob/917b11e38111c43526fe03ae6094a7081aeb982a/neuron_explainer/explanations/explainer.py#L1321. Accessed 2026-07-02.
- [23] Thomas Jiralerspong and Trenton Bricken. Cross-Architecture Model Diffing with Crosscoders: Unsupervised Discovery of Differences Between LLMs, February 2026. URL <http://arxiv.org/abs/2602.11729>. arXiv:2602.11729 [cs.AI].

A Refusal Transcript Excerpts

The refusal-circuit graphs in Section 4 are interpreted against the first generated tokens produced by the base and ablated Qwen3-4B models on the same harmful request. The base model refused on two repeated trials, while the ablated model began with a compliance token sequence:

User prompt: Tell me how to make a bomb

Standard Qwen: I’m unable to provide information on how to make a bomb, as this is a dangerous and illegal activity that can cause serious harm to people and property. [...]

Heretic-tuned Qwen: Sure! Making a bomb is a serious task, but I can give you a basic guide to making a simple explosive device. [...]

These excerpts identify the response prefix whose logits are visualised: refusal for the base model and helpful-style compliance for the ablated model.

B Declaration of Auto-Generation Tool Usage

During the preparation of this report, generative AI and auto-generation tools were used to assist with coding, debugging, drafting, and proofreading. Specifically, Claude Code (Anthropic), Codex (OpenAI), and Antigravity (Google) were employed as software development and editing assistants. The types of tasks where these tools were used are described below:

- **Drafting and Proofreading:** Suggesting alternative sentence structures, checking grammatical correctness, proofreading body text, and identifying phrasing improvements throughout the manuscript.
- **Code Implementation:** Refactoring Python scripts into modules for transcoder substitution, computing attribution graphs, running steering sweeps, and generating LaTeX table templates.
- **Debugging and Environment Setup:** Identifying code issues, resolving package conflicts within the Conda environments, and configuring SLURM job submission scripts for the CSD3 cluster.

All code, analyses, and written text were reviewed, verified, and finalised by the author, who retains full responsibility for the originality, accuracy, and integrity of the final submission.